

Arvin Zaker

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1 Educational Background

Doctor of Medicine | (August 2025 - Present)

University of British Columbia, Ottawa, ON

Honours Bachelor of Science - Translational and Molecular Medicine | (September 2019 - April 2023)

University of Ottawa, Ottawa, ON

- Major Research Project Supervisor: Dr. Arvind Mer
- Major Research Project Title: Comparative analysis of drug response metrics in patient-derived xenografts

2 Honours, Scholarships and Awards

NSERC - Undergraduate Student Research Awards (USRA) | (May 2023)

University of Ottawa, Ottawa, ON

- Competitive merit-based award. One of 88 awards available at the University of Ottawa.
- Selection based on quality and impact of the research proposal.
- Valued at \$7500 for 3 months

Work in Biomedical Research Scholarship | (May 2022)

University of Ottawa, Ottawa, ON

- Competitive merit-based award. Under 50% success rate.
- Selection based on academic performance and the quality of the research proposal.
- Valued at \$3500 for 3 months

TMM Summer Training program | (May 2022)

University of Ottawa, Ottawa, ON

- Competitive merit-based award. Selection based on academic performance and the quality of the research proposal.
- Valued at \$8900 for 3 months.

Undergraduate Research Opportunities (UROP) | (December 2020)

University of Ottawa, Ottawa, ON

- Competitive based award. Selection based on academic performance and the quality of the research proposal.
- Valued at \$1000 for 4 months.

Dean's Honour List & Merit Scholarship | (September 2019 - April 2023)

University of Ottawa, Ottawa, ON

- Merit-based award. Selection based on grade point average (GPA) in post-secondary studies.
- Valued at \$1000 per term (awarded every term).

3 Research Experience

Research Assistant – Medical Imaging AI

Feb 2026 – Present

Supervisor: Dr. Ilker Hacihaliloglu

Contribution:

- Applied deep learning models for medical image classification on 7 datasets totaling 40,000+ images.
- Performed radiomic feature extraction and model evaluation across multiple architectures to quantify bias in medical imaging models.
- Utilized high-performance computing for large-scale training and analysis.

Research Assistant - Epigenomics

March 2024 - Jan 2026

Supervisor: Dr. Arvind Mer (Primary Supervisor)

Contribution:

- Collaborators: Dr. Ben Tsang, Dr. Gilda Stefanelli, and Dr. Jean-Francois Couture.
- Perform epigenomic analysis of clinical exosomal DNA samples to identify differentially methylated regions and pathways.
- Created and optimized Cut&Run epigenomic analysis pipeline to explore the developmental impacts of neurological conditions in mice.
- Integrated and aggregated multi-omics data from 44 studies and databases.

Undergraduate Thesis

September 2022 - Feb 2024

Supervisor: Dr. Arvind Mer

Title: Comparative Analysis of Drug Response Metrics for Patient-Derived Xenograft

Contribution:

- Performed literature review to identify common methods of quantifying drug response in patient-derived xenografts (PDX).
- Developed a package to calculate 250 existing and novel drug response metrics in PDX.
- Performed biomarker discovery to identify transcriptional markers of drug sensitivity in chemotherapies.
- Created machine learning algorithms to predict drug response in clinical datasets.

Summer Research Assistant

January 2022 - August 2022

Supervisor: Dr. Arvind Mer (Primary Supervisor) and Dr. Damien D'Amours

Title: Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer

Contribution:

- Aggregated and combined data from 199 cancer studies to analyze the mutational effects of SMC5/6 DNA repair complex in a pan-cancer cohort.
- Utilized R programming language to analyze the functional impact of SMC5/6 mutation using transcriptomic data.
- Performed cancer phenotype and survival analysis to study the clinical implication of SMC5/6.

Student Research Assistant

September 2021 - April 2022

Supervisor: Dr. Adam Rudner

Title: Discovery and Characterization of Phage Arzan and Khorshid

Contribution:

- Performed environmental sampling, phage extraction, and phage amplification assays.
- Conducted DNA extraction and restriction digest analysis via gel electrophoresis.
- Discovered and annotated the genome of two novel phages, Arzan and Khorshid.

UROP Research Assistant

November 2020 - July 2022

Supervisor: Dr. Jean-Marc Renaud

Title: Role of Adenosine Receptors on Mice Skeletal Muscle Fatigue Kinetics.

Contribution:

- Designed experiments and set up research equipment for data collection.
- Performed experiments on mouse muscles using electrophysiological equipment.
- Performed antibody staining, fluorescent imaging, and analysis of muscle tissue in the Imiris program.

4 Publications, Presentations, and Abstracts

Publications

Asare-Werehene, M., **Zaker, A.**, Tripathi, S., Communal, L., Carmona, E., Mes-Masson, A.-M., Tsang, B. K., & Mer, A. (2025). Spotlight on nuclear PD-L1 in ovarian cancer chemoresistance: hidden but mighty. *Frontiers in Immunology*, 16. DOI: [10.3389/FIMMU.2025.1543529](https://doi.org/10.3389/FIMMU.2025.1543529)

Grégoire, S., Grégoire, J., Yang, Y., Capitani, S., Joshi, M., Sarvan, S., **Zaker, A.**, Ning, Z., Figeys, D., Ulrich, K., Brunzelle, J. S., Mer, A., & Couture, J.-F. (2024). Structural insights into an atypical histone binding mechanism by a PHD finger. *Structure*, 32(9), 1498-1506.e4. DOI: [10.1016/J.STR.2024.06.017](https://doi.org/10.1016/J.STR.2024.06.017)

Wu, G., **Zaker, A.**, Ebrahimi, A., Tripathi, S., & Mer, A. S. (2024). Text-mining-based feature selection for anticancer drug response prediction. *Bioinformatics Advances*, 4(1), vbae047. DOI: [10.1093/BIOADV/VBAE047](https://doi.org/10.1093/BIOADV/VBAE047)

Nasser, F., Gaudreau, A., Lubega, S., **Zaker, A.**, Xia, X., Mer, A. S., & D'Costa, V. M. (2024). Characterization of the diversity of type IV secretion system-encoding plasmids in *Acinetobacter*. *Emerging Microbes & Infections*, 13(1), 2320929. DOI: [10.1080/22221751.2024.2320929](https://doi.org/10.1080/22221751.2024.2320929)

Roy, S., **Zaker, A.**, Mer, A., & D'Amours, D. (2023). Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *NAR Cancer*, 5(3), zcad047. DOI: [10.1093/NARCAN/ZCAD047](https://doi.org/10.1093/NARCAN/ZCAD047)

Posters and presentations at scientific meetings

Roy, S., **Zaker, A.**, Mer, A., D'Amours, D. (2023). Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *Canadian Society for Molecular Biosciences (CSMB) International Conference*. Ottawa, ON. (poster and presentation)

Zaker, A., Mer, A. (2022). Comparative analysis of drug response metrics in patient-derived xenografts. *4th Annual University of Ottawa Faculty of Medicine Research Day*. Ottawa, ON. (oral presentation)

Friesen, R., Algharbi S., **Zaker, A.**, Mahdi, O., Gao, R., Ferri, O., Li, J., Wang, SY., Heffernan C., Featherstone, A., Radar, A., Gandelman, M., Chander N., Bancud, SE., Rege, I., Shiraam, R., Jung, D., Karunakaran, G., Sarakbi, R., Znamenski, E., Ristovski, M., Freitas, JD., McCarthy, L., Williams, EC., D'Ambrosio, L., Chan, K., Wheaton, K., Rudner, AD. (2022). Investigating Nucleotide-binding Proteins in Bacteriophage JohnDoe. *SEA Symposium*. Pittsburgh, PA. (abstract & presentation).

Featherstone, A., Radar, A., Salama, A., Tiukuvaara, S., Ferri, O., **Zaker, A.**, Algharbi, S., Mahdi, O., Setia, G., Friesen, R., Wang, GY., Li, J., Gao, R., Jung, D., Karunakaran, G., Znamenski, E., Ristovski, M., Sarakbi, R., Freitas, JD., McCarthy, L., Williams, EC., D'Ambrosio, L., Wheaton, K., Rudner AD. (2022). Give Us a Millet of Your Time: Dehusking the Singleton Bacteriophage Arzan. *SEA Symposium*. Pittsburgh, PA. (abstract & presentation)

Zaker, A., McCarthy, L., Jung, D., Karunakaran, G., Rudner, A. (2022). Discovery and characterization of bacteriophage Arzan. *Ottawa, ON*. (poster and presentation)

Zaker, A., McRae, C., Renaud, J.M. (2021). Role of Adenosine Receptors on mice skeletal muscle fatigue kinetics. *University of Ottawa UROP symposium*. Ottawa, ON. (abstract and poster).

5 Software Projects and Repositories

cut_run_pipeline_nix

September 2024

- Collaborative project with Dr. Stefanelli lab.
- Developed a pipeline for performing Cut&Run analysis and annotating the analysis outcomes.
- Successfully stacked multiple technologies to create a robust pipeline.
- Created a user-friendly guide for improving the utility of the pipeline for our collaborators.
- Improved the reproducibility of existing pipelines by utilizing Nix programming language.

text_features

March 2024

- Compared and analyzed the impact of various feature selection methods in drug response prediction models.
- Created scripts to be used in high-performance computing environment in the cloud and locally.
- Created large-scale visualizations to communicate abstract findings to a wide audience.

SMC5-6_complex

September, 2023

- Collaborative project with Dr. D'Amours lab.
- Aggregated pan-cancer dataset of over 65,000 patients from across 144 studies.
- Analyzed the mutational effect of the SMC5/6 protein complex and its subunits.
- Performed survival analysis to identify the SMC5/6-related risk factors to cancer prognosis.
- Identified large-scale genomic instability associated with SMC5/6 mutation in patient transcriptome.

vian-clin-sci

March 2023

- Reproduced the bioinformatics analysis performed by Viñas et al. (2020) [PMID: 32662516](#).
- Developed a comprehensive report and delivered an in-depth presentation of the re-analysis, incorporating detailed visualizations and narrative summaries.
- Optimized the analysis outcomes by integrating state-of-the-art bioinformatics tools to enhance data resolution and accuracy.

ATG16L1-MYPT1

November 2022

- Collaborative project with Dr. Ryan Russel lab.
- Performed a differential expression analysis of VHL and RB knockout mice samples.
- Identified the differentially expressed genes and pathways associated with knockouts.
- Verified findings using novel literature text mining tools.